

# A GARCH-Scaled Hierarchical Bayesian Model for Calibrated Daily Return Densities: Evidence from Large-Cap Technology Stocks, 2016–2025

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## Abstract

Forecasts of daily equity returns are usually evaluated on point accuracy, yet most practical uses—risk limits, position sizing, option pricing—require a predictive *distribution*. We develop and evaluate a hierarchical Bayesian model that supplies one. The specification treats a GARCH(1,1)- $t$  conditional standard deviation as a known observation scale, pools intercepts across assets, places a horseshoe prior on the mean coefficients so that predictor selection occurs inside the posterior, and uses Student- $t$  innovations with a scale calibrated on validation data. Estimation is by blocked Gibbs sampling with all conditionals in closed form; the sampler recovers known parameters on synthetic data and satisfies standard convergence diagnostics.

On a balanced panel of eight large-capitalisation United States technology stocks over 2016–2025, evaluated on a held-out window spanning the 2022–2023 monetary tightening cycle, the model attains the best logarithmic score and continuous ranked probability score among all specifications considered, improving on a ridge-plus-GARCH benchmark by 0.126 nats ( $p < 0.00001$ ), and reduces mean absolute interval-coverage error to 0.022 from 0.030. Partial pooling delivers a further practical benefit: for an asset excluded from estimation entirely, the model still produces intervals attaining 87% coverage at the 90% nominal level, with a logarithmic score within 0.005 nats of the same model estimated with that asset included.

Decomposing the gain in predictive density over a null forecast shows where the value originates: the conditional variance contributes 54% and the shape of the predictive distribution a further 45%, while the conditional mean contributes 0.7%. Consistent with the out-of-sample evidence of Welch and Goyal (2008) and Timmermann (2008), no specification improves on a per-ticker historical average for the conditional mean, which is precisely why modelling effort at this horizon is better directed at the second moment and the distributional shape. We additionally document a silent failure mode: standardising non-stationary macroeconomic levels on a fixed estimation origin sends held-out inputs to 8.7 training standard deviations and produces out-of-sample  $R^2$  as low as  $-10\,425\%$  with no error raised. All code, audit scripts and pinned dependencies are released.

**Keywords:** density forecasting; forecast calibration; hierarchical Bayesian models; GARCH; cross-asset transfer

## 1 Introduction

Between 2016 and 2025, large-capitalisation United States technology equities traded through several distinct macro-financial regimes: an extended period of near-zero policy rates, the COVID-19

market dislocation of March 2020, the most rapid monetary tightening cycle since the early 1980s beginning in 2022, and the partial normalisation of rates that followed. These episodes produced pronounced volatility clustering, heavy-tailed daily returns, and repeated shifts in the joint distribution of returns and macroeconomic state variables—features long documented as stylised facts of asset returns [Cont, 2001]. Such conditions are hostile to forecasting models estimated under an assumption of covariance stationarity, and they reproduce at the level of a portfolio a persistent finding of the empirical finance literature: predictive relations that appear stable in sample frequently deteriorate out of sample [Welch and Goyal, 2008, Timmermann, 2008]. As West [2020] argues, structure uncertainty then becomes as consequential as parameter uncertainty, since the analyst must decide not only what the coefficients are but which model, if any, remains adequate as the regime changes.

There are sound theoretical grounds for conditioning return forecasts on macro-financial information. Multifactor asset pricing theory implies that innovations in systematic state variables are priced in equity returns [Chen et al., 1986]. Empirically, unanticipated changes in the federal funds rate induce large and statistically significant equity market responses [Bernanke and Kuttner, 2005]; option-implied volatility, summarised by the CBOE Volatility Index, is contemporaneously and negatively associated with returns [Whaley, 2000]; and comovement with the broad market index remains the dominant source of systematic variation in individual stock returns. Recent data-driven studies confirm that volatility and uncertainty measures rank among the most influential predictors of index returns [Latif et al., 2025], and that volatility transmission across markets is itself time varying [Agrawal et al., 2026]. The unresolved methodological question is therefore not whether such variables carry information, but how they should enter the forecasting model and how the resulting predictive uncertainty should be quantified.

This paper approaches that question constructively. Most practical uses of a return forecast—setting a risk limit, sizing a position, pricing an option—require a predictive *distribution* rather than a point estimate, yet the forecasting literature is dominated by evaluations of conditional-mean accuracy. We develop a model that supplies a calibrated distribution, and we establish which of its components does the work.

The specification is a hierarchical Bayesian regression with four features. It treats a GARCH(1,1)- $t$  conditional standard deviation as a *known* observation-level scale, so the model does not have to relearn volatility clustering and a single scalar measures whether that scale is correctly sized. It pools intercepts across assets, which allows an asset absent from estimation to be forecast by integrating its intercept over the population distribution. It places a horseshoe prior [Carvalho et al., 2010] on the mean coefficients, so that predictor selection occurs inside the posterior with uncertainty attached rather than as a screening step beforehand. And it uses Student- $t$  innovations with a scale calibrated on validation data, which separates tail behaviour from scale. Estimation is by blocked Gibbs sampling with every conditional in closed form.

The evaluation is a controlled comparison on a balanced panel of eight large-cap technology stocks over 2016–2025, with the effective federal funds rate, the ten-year Treasury yield, the VIX and a market proxy as macro-financial conditioning information. Competing specifications—naïve baselines, pooled ridge, a pooled random forest, three GARCH-family volatility models and four sequence architectures—are estimated on an identical feature set and scored on a strictly held-out window spanning the tightening cycle and its aftermath. Point accuracy is assessed by root mean squared error and out-of-sample  $R^2$ ; probabilistic adequacy by logarithmic score, continuous ranked probability score and interval calibration. Every comparison is tested by Diebold–Mariano rather than ranked.

We report four findings. First, the proposed model attains the best logarithmic score and CRPS among all specifications considered, improving on a ridge-plus-GARCH benchmark by 0.126 nats

( $p < 0.00001$ ), and reduces mean absolute interval-coverage error to 0.022 from 0.030. Second, partial pooling yields calibrated intervals for an asset with no estimation history at all, attaining 87% coverage at the 90% nominal level and a logarithmic score within 0.005 nats of the same model estimated with that asset included. Third, decomposing the gain in predictive density over a null forecast locates its source: the conditional variance contributes 54% and the shape of the predictive distribution a further 45%, while the conditional mean contributes 0.7%—detectable at  $p = 0.029$  and economically negligible. Fourth, consistent with Welch and Goyal [2008] and Timmermann [2008], no specification improves on a per-ticker historical average for the conditional mean, a result that survives disaggregation to individual assets and correction for multiple testing, and which is precisely why modelling effort at this horizon is better directed at the second moment and the distributional shape.

We also document a failure mode that is silent rather than merely severe. Standardising non-stationary macroeconomic levels on a fixed estimation origin sends held-out inputs far outside the range seen in estimation, and every model with an unbounded output head extrapolates along them, producing out-of-sample  $R^2$  as low as  $-10\,425\%$  with no exception raised and no implausible training diagnostic. Because the defect is detectable only by auditing predictor distributions across splits and the scale of the predictions themselves, both audits are released with the code.

The remainder of the paper reviews the relevant literature, describes the data and methodology, reports the empirical results, and concludes with recommendations for practice.

## 2 Literature Review

### 2.1 Classical time-series models and the limits of return predictability

The autoregressive integrated moving average framework of Box and Jenkins [1970] remains the canonical linear model for the conditional mean of a univariate series and the natural benchmark against which richer specifications are judged. Its known limitation in financial applications lies in the second moment: daily returns exhibit conditional heteroskedasticity, which motivated the ARCH model of Engle [1982] and its GARCH generalisation [Bollerslev, 1986], and which implies that a correctly specified linear conditional-mean model may still misstate predictive uncertainty by a wide margin. A second limitation concerns the conditional mean itself. In a comprehensive out-of-sample audit, Welch and Goyal [2008] found that most proposed return predictors fail to improve upon the historical average, and Timmermann [2008] characterised return predictability as local and episodic—present in short-lived pockets rather than as a stable population property. Forecast combination offers partial insurance against this instability [Rapach et al., 2010], but the collective evidence establishes a demanding baseline: any gain claimed for a more elaborate model must survive genuine out-of-sample evaluation over heterogeneous regimes.

### 2.2 Machine learning approaches

Machine learning methods relax the linearity restriction at the cost of a larger effective parameter space. Random forests [Breiman, 2001] control overfitting through bootstrap aggregation of decorrelated trees and provide internal measures of predictor importance, properties that make them a common nonlinear benchmark in return forecasting. In the most systematic comparison to date, Gu et al. [2020] showed that regression trees and neural networks yield economically significant improvements in measuring equity risk premia, attributing the gains chiefly to nonlinear predictor interactions that linear methods miss. For sequence models, the LSTM architecture [Hochreiter and Schmidhuber, 1997] was shown by Fischer and Krauss [2018] to outperform random forests

and logistic regression in classifying daily S&P 500 constituent returns, although the documented profitability of the strategy declined in later subperiods—an instability consistent with the pocketed predictability described above. Subsequent applied work has extended these architectures with macroeconomic inputs and hybrid convolutional–recurrent designs [Latif et al., 2025]. Two limitations recur throughout this literature: the fitted models are difficult to interpret, and their outputs are point predictions unaccompanied by a distributional statement.

A third limitation is less often discussed and is central to this paper. Because these models are typically reported at a single regularisation setting chosen by the author rather than selected on held-out data, it is difficult to distinguish a model that has found signal from one that has fitted noise. Section ?? shows that the distinction is decisive here.

### 2.3 Bayesian inference and uncertainty quantification

The Bayesian literature addresses the distributional limitation directly. West [2020] surveys multivariate Bayesian forecasting with emphasis on scalability and structure uncertainty, arguing that decision-relevant forecasts require the full predictive distribution rather than a summary of its centre. Qiu et al. [2018] develop the multivariate Bayesian structural time series model, in which local linear trends, seasonal components and spike-and-slab selection over a candidate pool of predictors jointly capture contemporaneous correlation across target series while guarding against overfitting. In the return-predictability setting specifically, Bayesian treatments of model uncertainty—averaging over predictor subsets rather than conditioning on a single specification—were shown by Avramov [2002] and Cremers [2002] to improve out-of-sample performance precisely because no individual model is reliably correct. Closer to the present study, Chandra and He [2021] apply Bayesian neural networks to multi-step stock price forecasting and report usable predictive intervals even through the COVID-19 volatility spike, while Wong et al. [2024] show that standard uncertainty-quantification schemes for neural networks miscalibrate under volatility clustering, underscoring that calibration must be verified rather than assumed. Proper scoring rules [Gneiting and Raftery, 2007] and tests of equal predictive accuracy [Diebold and Mariano, 1995] supply the formal apparatus for such verification, and this paper applies both throughout.

For coefficient shrinkage we adopt the horseshoe prior of Carvalho et al. [2010], the continuous analogue of the spike-and-slab selection used by Qiu et al. [2018], sampled by the auxiliary-variable scheme of Makalic and Schmidt [2016].

### 2.4 Distribution shift and the fragility of fixed-origin evaluation

A strand of the machine learning literature that has had limited uptake in empirical finance concerns dataset shift: the case in which the joint distribution of predictors and target differs between estimation and deployment [Quiñonero-Candela et al., 2009]. The concern is directly relevant here, because the macro-financial variables most commonly used to condition equity forecasts—an index level, a policy rate, a yield—are precisely the series whose means move permanently across a monetary regime change. Standardising such a series on a fixed estimation window and applying the transformation to a later window produces inputs far outside the range the estimator has seen, and any model with an unbounded output head extrapolates along them. Section ?? documents the consequences and the repair.

### 2.5 Hybrid architectures and the remaining gap

A growing strand seeks to combine the representational capacity of recurrent networks with the distributional discipline of econometric volatility models. Kim and Won [2018] integrate LSTM

networks with multiple GARCH-type specifications and report improved index volatility forecasts relative to either component alone, and Wang et al. [2022] document analogous gains for a combined LSTM–GARCH-family model in a highly volatile price series. These hybrids, however, typically attach a volatility filter to a point-forecasting network rather than deliver a coherent posterior predictive distribution, and published comparisons rarely hold the information set fixed across model families.

The present study is positioned in that gap. It evaluates classical, machine learning and Bayesian models on a single panel with an identical macro-financial predictor set and an identical regime-spanning out-of-sample window, and it scores them on both point accuracy and the calibration of their stated uncertainty. To the extent that the literature reviewed above establishes weak and unstable conditional-mean predictability alongside strong and persistent structure in conditional variances, the comparison speaks to a practical question left open by prior work: whether the principal contribution of macroeconomic drivers and Bayesian inference at the daily horizon lies in sharper point forecasts or in more honest predictive distributions. The decomposition in Section 4.2 answers it quantitatively.

### 3 Data and Methodology

#### 3.1 Sample and splits

The sample is a balanced daily panel of eight large-capitalisation United States technology equities—AAPL, MSFT, AMZN, GOOGL, META, NVDA, TSLA and ORCL—from 2 March 2016 to 29 December 2025. Adjusted closing prices are obtained from Yahoo Finance and macro-financial series from the Federal Reserve Economic Data service: the effective federal funds rate (DFF), the ten-year Treasury constant maturity yield (DGS10), the S&P 500 index (SP500) and the CBOE volatility index (VIX). The market proxy is the SPY exchange-traded fund.

The forecast target is the one-day-ahead log return. For ticker  $i$ , a panel row is keyed by date  $t$  and its target is

$$y_{i,t} = \log P_{i,t+1} - \log P_{i,t}, \quad (1)$$

realised on  $t + 1$ . Every predictor is observable at the close of  $t$ , so no input uses information from the period being forecast.

The panel is divided chronologically into three blocks, with no shuffling at any stage:

| Split      | Dates                    | Rows  |
|------------|--------------------------|-------|
| Training   | 2016-03-02 to 2019-12-30 | 7 720 |
| Validation | 2019-12-31 to 2022-12-29 | 6 048 |
| Test       | 2022-12-30 to 2025-12-29 | 6 008 |

The split is deliberately demanding. The training block spans a low-rate, low-volatility regime; the validation block contains the COVID-19 dislocation; and the test block spans the 2022–2023 tightening cycle and its aftermath. Any relation that survives is required to survive a change in monetary regime.

#### 3.2 Feature dictionary and stationarity

Model inputs are declared explicitly rather than inferred from column type. This is a substantive design choice, not bookkeeping. Selecting every numeric column admits level series—an index level,

a policy rate—whose mean moves permanently between the training and test blocks. Standardised on training moments and applied to the test block, such a series arrives at a magnitude the estimator has never seen, and any model with an unbounded linear output extrapolates along it. Section ?? reports the consequences directly.

Membership is therefore fixed in advance, and each candidate is classified by an augmented Dickey–Fuller test. Three series are non-stationary over the sample: the S&P 500 level ( $p = 0.99$ ), the effective federal funds rate ( $p = 0.83$ ) and the ten-year yield ( $p = 0.80$ ). All three are excluded and replaced by their first differences or by a stationary counterpart. The retained set comprises fourteen predictors: six own-return lags, three trailing realised volatilities, the market return, the log VIX and its change, and the daily changes in the federal funds rate and the ten-year yield. Hybrid specifications add one further input, the GARCH conditional standard deviation of Section 3.4, so base and hybrid models differ by exactly one column. Table 1 lists every candidate, retained or excluded, with the reason for exclusion.

Two redundancies are also removed. One column duplicated another exactly, and the S&P 500 return correlated 0.9985 with the market proxy return; retaining both would inflate collinearity and render the Bayesian coefficient posteriors uninterpretable.

### 3.3 Evaluation protocols

Two protocols are used throughout, and the difference between them is itself a result.

Under the *fixed-origin* protocol a model is estimated once on the training block, and the fitted model together with its feature scaler is frozen and applied to the whole test block. This is the conventional design and is appropriate when predictors are stationary.

Under the *expanding-window* protocol the model walks forward month by month. At each fold it is refitted on every observation whose target date falls strictly before the fold start, and it forecasts that month only; the scaler is refitted inside the fold along with the model. This is what a forecaster could have executed in real time.

Both protocols score the identical 6 008 test observations, so their metrics are directly comparable, and across all 36 folds no training observation reaches into the month being scored.

### 3.4 Conditional volatility

Three conditional volatility specifications are estimated per ticker on returns in percent: GARCH(1,1), GJR-GARCH(1,1) and EGARCH(1,1), each with Student- $t$  innovations. Parameters are estimated on the training block alone. They are then held fixed and the recursion is filtered forward through the validation and test blocks, so the stored quantity is a genuine one-step-ahead forecast  $\sigma_{i,t+1|t}$  that uses only information through  $t$ . The value recorded at date  $t$  therefore forecasts the same period as the target in (1).

Every fit is checked for optimiser convergence, finite parameters and a conditional standard deviation within a plausible range before it is written; the pipeline aborts otherwise. All twenty-four fits (eight tickers, three specifications) satisfy these conditions, with conditional standard deviations between 0.6% and 13.8% per day against a realised return standard deviation of 2.5%. Per-fit parameters, log-likelihoods and AIC values appear in Appendix Table ??.

### 3.5 Hierarchical Bayesian model

The central specification pools information across tickers while treating the GARCH forecast as a known observation-level scale. For ticker  $i$  on day  $t$ , with  $\mathbf{x}_{i,t}$  the standardised predictor vector,

$$y_{i,t} \mid \alpha_i, \boldsymbol{\beta}, s^2 \sim t_\nu \left( \alpha_i + \mathbf{x}_{i,t}^\top \boldsymbol{\beta}, s^2 \sigma_{i,t+1|t}^2 \right), \quad (2)$$

$$\alpha_i \mid \mu_\alpha, \tau_\alpha^2 \sim \mathcal{N}(\mu_\alpha, \tau_\alpha^2), \quad (3)$$

$$\beta_j \mid \lambda_j, \tau_\beta \sim \mathcal{N}(0, \lambda_j^2 \tau_\beta^2), \quad \lambda_j \sim \mathcal{C}^+(0, 1), \quad \tau_\beta \sim \mathcal{C}^+(0, 1). \quad (4)$$

Three features of this specification carry the analysis.

The GARCH forecast enters (2) as a known scale, so the model does not relearn volatility clustering and  $s^2$  is a single scalar measuring whether that scale is correctly sized overall. A value near one means the volatility model is already calibrated.

Equation (3) is the partial-pooling layer. Because  $\alpha_i$  is drawn from a population distribution, a ticker absent from the training sample can still be forecast by integrating  $\alpha$  over that distribution, which adds  $\tau_\alpha^2$  to the predictive variance. Section 4.3 exploits this. Following Gelman [2006],  $\tau_\alpha$  receives a half-Cauchy prior rather than a conventional inverse-gamma: with only eight groups the between-ticker standard deviation is weakly identified, and a diffuse inverse-gamma prior largely reproduces its own mode.

Equation (4) is the horseshoe prior of Carvalho et al. [2010]. It replaces the more common practice of pre-screening predictors by their correlation with the target. That practice is unsafe in this setting for a specific reason: a trending series correlates with almost any target over a multi-year window, so a correlation screen preferentially selects exactly the non-stationary levels that Section 3.2 excludes. The horseshoe retains every predictor and shrinks weak coefficients continuously toward zero while leaving strong ones nearly unshrunk, so selection occurs inside the posterior and carries uncertainty. It is the continuous analogue of the spike-and-slab selection of Qiu et al. [2018].

Student- $t$  innovations are represented as a scale mixture of normals,  $y \sim \mathcal{N}(\mu, s^2 \sigma^2 / \omega)$  with  $\omega \sim \text{Gamma}(\nu/2, \nu/2)$ , which recovers (2) marginally and keeps every conditional in closed form.

Inference is by blocked Gibbs sampling. All conditionals are conjugate, including the horseshoe scales, via the auxiliary-variable representation of Makalic and Schmidt [2016]; the degrees of freedom  $\nu$  are drawn by griddy Gibbs. Four chains are run from dispersed starting points for 40 000 iterations, discarding the first 20 000 and thinning by four, giving 20 000 retained draws. Convergence is assessed by rank-normalised  $\hat{R}$  and effective sample size [Vehtari et al., 2021]: across 27 monitored quantities the largest  $\hat{R}$  is 1.0033 and the smallest bulk effective sample size is 1 703.

The sampler is validated on synthetic data in which three of twelve coefficients are non-zero. It recovers all three signal coefficients with credible intervals excluding zero, produces no false positives among the nine null coefficients, and recovers the observation scale and the between-group standard deviation.

### 3.6 Predictive distributions and scale calibration

Predictive means and scales use the full posterior rather than plug-in point estimates, so parameter uncertainty enters the interval through the law of total variance.

One calibration step is applied. GARCH parameters are frozen at the end of the training block, and the training block is calmer than what follows: measured as the standard deviation of  $y/\sigma$ , the scale the data require is 1.002 on the training block but 1.118 on validation and 1.117 on test. A scale fitted to training data alone therefore understates later volatility by roughly twelve percent. A

single scalar is estimated on the *validation* block and applied to the test block; no test observation informs it.

The matching statistic matters. Because the validation block contains the COVID-19 dislocation, a variance match is dominated by a few extreme days, which is behaviour the Student- $t$  tails are meant to absorb rather than the scale. Matching the interquartile range targets the body of the distribution, which is what the scale parameter governs, and yields a factor of 1.195 against 1.220 for the variance match.

### 3.7 Scoring rules and inference

Point accuracy is reported as root mean squared error, mean absolute error, directional accuracy and out-of-sample  $R^2$  against a stated reference forecast. The reference is the per-ticker mean computed under the same protocol as the model being scored, so it never enjoys an information advantage.

Probabilistic accuracy is reported as the mean logarithmic score, the continuous ranked probability score and empirical coverage at the 50%, 80%, 90% and 95% levels [Gneiting and Raftery, 2007]. All three are reported because they can disagree: a model may be better calibrated while being less sharp. Calibration is assessed additionally by the probability integral transform, whose values are uniform under a correctly specified predictive distribution, tested formally by a Kolmogorov–Smirnov statistic.

Volatility forecasts are compared under QLIKE against the squared return as the realised-variance proxy. QLIKE is chosen because it preserves the ranking a perfect proxy would give even when the available proxy is noisy [Patton, 2011].

All comparisons are tested rather than asserted. Equal predictive accuracy is assessed by the Diebold and Mariano [1995] test on per-observation loss differentials, using a Newey–West heteroskedasticity- and autocorrelation-consistent variance because daily loss differentials are serially correlated, together with the small-sample correction of Harvey et al. [1997]. Where a family of tests is reported—one per model–ticker pair—significance is additionally assessed after Benjamini–Hochberg and Bonferroni correction, since at conventional levels a handful of rejections among dozens of tests is what chance alone produces.

## 4 Results and Discussion

All results are computed on the same held-out block of 6 008 observations covering 2023–2025, against the same benchmarks, and every comparison is tested by Diebold–Mariano rather than ranked by point estimate.

### 4.1 Density forecast accuracy and calibration

Table 4 reports the principal result. The proposed model attains the best logarithmic score and the best continuous ranked probability score among all specifications considered, and the margins are large relative to their standard errors.

Against a ridge-plus-GARCH benchmark—itsself a strong competitor, since it carries the same conditional variance and a tuned conditional mean—the model improves the mean log predictive density from 2.3516 to 2.4777, a gain of 0.126 nats with  $DM = -5.51$  ( $p < 0.00001$ ). Against a zero-mean forecast with the same GARCH scale the gain is 0.128 nats ( $p < 0.00001$ ), and against a constant-variance Gaussian predictive it is 0.281 nats ( $p < 0.00001$ ). CRPS orders the specifications identically, at 0.01214 for the proposed model against 0.01220 and 0.01246 for the two benchmarks.



Calibration improves alongside sharpness, which is not automatic: a model can buy log score by concentrating probability where the data happen to fall. Figure 2 plots empirical coverage against nominal level. The proposed model attains 0.536, 0.825, 0.918 and 0.958 against the 50%, 80%, 90% and 95% levels, a mean absolute coverage error of 0.022. The Gaussian alternative attains 0.572, 0.834, 0.909 and 0.943, a mean absolute error of 0.030, and its error is concentrated in the body of the distribution: it over-covers the 50% interval by more than seven percentage points.

Figure 3 shows why. The Gaussian probability integral transform is visibly hump-shaped, with excess mass near the centre—the signature of a thin-tailed density fitted to fat-tailed returns, which widens everywhere to accommodate occasional large moves. Student- $t$  innovations with a validation-calibrated scale flatten it substantially, reducing the Kolmogorov–Smirnov statistic against uniformity from 0.0517 to 0.0206.

We state the limit of this result plainly. The Student- $t$  specification more than halves the KS statistic but still rejects uniformity at  $p = 0.012$ . Calibration is materially improved, not achieved, and Section 5 treats the residual misspecification as open.

## 4.2 Where the predictive gain originates

The model combines three ingredients, and it is worth establishing which of them carries the improvement. A predictive distribution has a location, a scale and a shape; Figure 5 adds one at a time to a null forecast and scores each rung on the same sample.

| Addition                        | Gain (nats) | Share | DM $p$    |
|---------------------------------|-------------|-------|-----------|
| Conditional variance (GARCH)    | +0.1528     | 54.4% | < 0.00001 |
| Conditional mean (ridge)        | +0.0020     | 0.7%  | 0.0293    |
| Fat tails and scale calibration | +0.1261     | 44.9% | < 0.00001 |
| Total                           | +0.2808     | 100%  |           |

Two components account for essentially all of the gain. The conditional variance supplies 54.4% and the shape of the predictive distribution a further 44.9%; getting the shape right is worth nearly as much as getting the scale right, and neither is visible to an evaluation confined to point forecasts.

The conditional mean supplies 0.7%. That contribution is statistically detectable at  $p = 0.029$  and economically trivial, and the two statements are not in tension: with 6 008 observations a Diebold–Mariano test resolves differences far below any that would affect a decision. We report both because a study quoting only the  $p$ -value would present this as evidence that macroeconomic conditioning materially improves density forecasts, which it does not.

This decomposition is the empirical justification for the model’s design. It is why the specification devotes its structure to the scale—a GARCH forecast entering as a known observation scale, Student- $t$  innovations, a calibrated scale factor—and treats the conditional mean parsimoniously, with a horseshoe prior that shrinks weak coefficients rather than a flexible functional form.

## 4.3 Forecasting an asset with no estimation history

Partial pooling delivers a capability that per-asset models cannot. Because intercepts are drawn from a population distribution, an asset absent from estimation can still be forecast by integrating its intercept over that distribution, which adds  $\tau_\alpha^2$  to the predictive variance.

Table 5 reports leave-one-ticker-out performance. An asset entirely excluded from estimation attains a mean log score of 2.4727 against 2.4777 for the same model estimated with that asset included—a gap of 0.005 nats—and retains 87% coverage at the 90% nominal level. Per ticker the

log score ranges from 1.896 for TSLA, the most volatile asset in the panel, to 2.862 for MSFT, the least.

Revealing the held-out asset’s own history adds little. Across  $k \in \{0, 20, 60, 120, 250\}$  trading days the mean log score moves from 2.4727 to 2.4722 and RMSE is unchanged to five decimal places, while the posterior demonstrably shifts between settings. The interpretation is that these assets are close to exchangeable for forecasting purposes: the between-ticker intercept variance is small relative to observation noise, so the population layer supplies almost everything.

For practice this is the more useful finding. A newly listed or thinly documented large-cap technology stock can be assigned a calibrated predictive distribution immediately, without waiting to accumulate an estimation sample.

#### 4.4 Which predictors survive shrinkage

Figure 4 reports the horseshoe shrinkage weight for each predictor. One of fourteen has a 95% credible interval excluding zero: the daily change in the effective federal funds rate, at 0.00063 with interval  $[0.00028, 0.00097]$  and shrinkage weight 0.89. The market return and the change in the VIX follow at weights 0.67 and 0.56, with intervals covering zero, and every own-return lag is shrunk toward zero.

The sign and direction are consistent with the monetary-policy transmission documented by Bernanke and Kuttner [2005]. The magnitude is small: roughly six basis points of next-day return per standard-deviation change in the policy rate, against a daily return standard deviation of 248 basis points. This is a statement about which conditioning variable retains any weight once the others compete for it, not a tradable signal.

Placing selection inside the posterior also avoids a specific hazard. Screening predictors by their correlation with the target on the estimation sample—common practice, and used in an earlier version of this work—preferentially selects trending series, because a trending regressor correlates with almost any target over a multi-year window. Those are precisely the series Section 4.7 shows to be destructive.

#### 4.5 The conditional variance

The GARCH component that supplies the model’s scale is itself worth validating. Table 3 reports QLIKE for three conditional specifications and a 20-day rolling-variance benchmark, all scored on an identical sample.

All three conditional models beat the benchmark decisively: GARCH(1,1)- $t$  at DM =  $-4.33$  ( $p = 0.00002$ ), GJR at  $-3.48$  ( $p = 0.0005$ ) and EGARCH at  $-3.32$  ( $p = 0.0009$ ).

Among the conditional specifications the differences are largely not significant. GARCH attains the best QLIKE at  $-6.536$ , but against EGARCH the difference carries  $p = 0.325$  and against GJR it sits on the boundary at  $p = 0.0499$ ; GJR against EGARCH gives  $p = 0.687$ . The asymmetric extensions, motivated by the leverage effect [Nelson, 1991, Glosten et al., 1993], deliver no measurable improvement for these assets over this window. The symmetric specification is therefore retained on parsimony grounds, and the choice is documented rather than assumed.

#### 4.6 Point accuracy

For completeness we report point accuracy, though it is not where the model is intended to compete.

No specification improves on a per-ticker historical average. Under the fixed-origin protocol pooled ridge attains an out-of-sample  $R^2$  of  $-0.11\%$  and a pooled random forest  $-0.34\%$ ; four sequence architectures, each permitted to select its own regularisation on held-out data, attain

between  $-0.04\%$  and  $-0.09\%$  and converge on forecasts whose standard deviation is zero to three decimal places. Diebold–Mariano tests against the benchmark return  $p$ -values between 0.13 and 0.51. Table 2 disaggregates by asset: across 48 model–ticker tests, one rejects at the uncorrected 5% level, fewer than the 2.4 that chance would produce, and it survives neither Benjamini–Hochberg [Benjamini and Hochberg, 1995] nor Bonferroni correction.

This is the expected result rather than a surprising one. It reproduces, on a different sample and with a wider model set, the out-of-sample evidence of Welch and Goyal [2008] and the episodic predictability described by Timmermann [2008]. Its role here is to motivate the paper’s design: if the conditional mean carries no exploitable signal at the daily horizon, then a forecasting effort evaluated only on point accuracy has almost nothing to report, and the predictive distribution is where the remaining value lies.

The sequence-model result deserves one further remark, because it is stronger than underperformance. Given free choice of regularisation on validation data, all four architectures select the strongest available penalty and collapse to a constant—they decline to use the predictors at all. The selected penalty sits at the top of the search grid, which would ordinarily indicate too narrow a search; here it does not, because forecast dispersion has already reached zero and no larger penalty can change it. The finding is conditional on this information set, decade and asset set, and is not a claim about market efficiency in general.

#### 4.7 A silent failure mode in fixed-origin standardisation

The final result is methodological, and we report it because the failure it describes produces no diagnostic of its own.

An earlier version of this pipeline selected model inputs by column type, which admitted three non-stationary level series: the S&P 500 index level, the effective federal funds rate and the ten-year yield. Standardised on 2016–2019 moments and applied to 2023–2025, the index level arrives at  $z = +8.7$ , with 100% of test observations beyond three training standard deviations; the policy rate arrives at  $z = +4.7$  and the ten-year yield at  $z = +3.9$  (Figure 1a).

Every model with an unbounded output head extrapolated along those axes. An LSTM’s test predictions correlate with the S&P 500 level at  $r = 0.88$  and imply a +12% return every trading day for three years. Sixteen of twenty-one saved models failed a prediction sanity check. Throughout, training error remained normal, no exception was raised, and metric files were written as usual.

Replacing the levels with stationary counterparts moves mean out-of-sample  $R^2$  across the eight assets from  $-6.23\%$  to  $-0.19\%$  for ridge, from  $-26.16\%$  to  $-0.47\%$  for the random forest and from  $-24.58\%$  to  $+0.13\%$  for the hierarchical model (Figure 1b). For the sequence models the shift is from per-ticker means between  $-638\%$  and  $-10\,425\%$  to between  $-0.04\%$  and  $-0.09\%$ .

Two lessons follow, both practical. Fitting a scaler once and freezing it is safe only for stationary predictors; where that cannot be guaranteed, the expanding-window protocol, which refits the scaler inside each fold, is the appropriate default, and on the uncorrected feature set it alone moves fixed-split ridge from  $-0.044$  to  $-0.013$  in out-of-sample  $R^2$ . More importantly, the failure is detectable only by auditing the distribution of predictors across splits and the scale of the predictions themselves. Both audits are released with the code, and both exit with a non-zero status when they detect the condition.

## 5 Conclusion and Recommendations

This study developed and evaluated a hierarchical Bayesian model for the predictive distribution of daily large-capitalisation technology equity returns, and established which of its components carries

the improvement.

The model performs well on the criterion it was built for. It attains the best logarithmic score and continuous ranked probability score among the specifications considered, improving on a ridge-plus-GARCH benchmark by 0.126 nats ( $p < 0.00001$ ), and reduces mean absolute interval-coverage error to 0.022 from 0.030. Partial pooling supplies a capability that per-asset models cannot: for an asset excluded from estimation entirely, the model still produces intervals attaining 87% coverage at the 90% nominal level, with a logarithmic score within 0.005 nats of the same model fitted with that asset included.

Decomposing the gain over a null forecast locates its source. The conditional variance contributes 54% and the shape of the predictive distribution a further 45%, while the conditional mean contributes 0.7%—statistically detectable and economically negligible. The last figure is consistent with the out-of-sample evidence of Welch and Goyal [2008] and the episodic predictability of Timmermann [2008], which our own point-accuracy results reproduce: no specification, including four sequence architectures permitted to select their own regularisation, improves on a per-ticker historical average. That is the empirical reason the model devotes its structure to the scale and the shape rather than to the mean.

## 5.1 Recommendations for practice

**Spend modelling effort on the second moment.** At the daily horizon the return of effort in conditional-mean modelling is close to zero, while conditional-variance modelling yields large and testable gains. A practitioner choosing where to allocate attention should not divide it evenly.

**Report calibration, not only accuracy.** Every conclusion above that matters is invisible to RMSE. The four rungs of Section 4.2 differ by less than 0.4% in RMSE and by 0.28 nats in log score. A study reporting only point accuracy would conclude that nothing distinguishes them.

**Test differences rather than ranking them.** Several apparent orderings in this study dissolve under testing: EGARCH’s QLIKE advantage over GARCH ( $p = 0.33$ ), the single per-ticker rejection that fails multiplicity correction, and—in an earlier version of this pipeline—an entire ranking of volatility models produced by a diverged fit.

**Separate statistical from economic significance.** The conditional mean’s contribution is real at  $p = 0.029$  and negligible at 0.7% of the total. With several thousand observations these two facts coexist routinely, and reporting only the first would misrepresent the finding.

**Declare features; never infer them from column type.** The single most consequential defect in the earlier version of this work was a feature-selection rule that admitted any numeric column. It produced no error, no warning and no implausible training diagnostic, and it invalidated every out-of-sample number in the study.

## 5.2 Limitations

The scope is eight assets in one sector over one decade. The between-ticker variance is identified by eight groups, which is few, and the transfer result of Section 4.3 should be read as evidence about large-capitalisation technology equities rather than about equities generally.

GARCH parameters are estimated once on the training block and then held fixed. This is conservative and avoids look-ahead, but Section ?? shows it also understates later volatility by roughly twelve percent. Rolling re-estimation is the obvious extension and is not attempted here.

Calibration is improved but not achieved: PIT uniformity is still rejected at  $p = 0.012$ . Candidate explanations include remaining asymmetry in the conditional distribution, time variation in

the degrees of freedom, and the single scalar used for scale calibration where a time-varying factor may be warranted.

The realised-variance proxy is the squared daily return. QLIKE is robust to proxy noise [Patton, 2011], but intraday data would support sharper volatility comparisons than are possible here.

Finally, the study conditions on one information set. The absence of conditional-mean predictability is a statement about these fourteen predictors over this window, and is consistent with, rather than a demonstration of, the episodic predictability described by Timmermann [2008].

### 5.3 Extensions

Three directions follow directly. Rolling GARCH re-estimation would test whether the scale miscalibration is an artefact of frozen parameters, and would make the calibration step unnecessary if so. A skewed or time-varying-shape predictive distribution would address the residual PIT non-uniformity. And extending the panel across sectors would test whether the near-exchangeability found here is a property of large-capitalisation technology equities specifically, or a more general feature of the asset class.

### 5.4 Reproducibility

All data are public: prices from Yahoo Finance, macro-financial series from the Federal Reserve Economic Data service. The full pipeline—panel construction, feature dictionary, volatility estimation, the Bayesian sampler, both evaluation protocols, all scoring rules and every figure and table in this paper—is released with pinned dependencies. Two audit scripts are included that exit non-zero when they detect the predictor-drift and prediction-scale failures described in Section ??, so that the defect which invalidated the earlier analysis cannot recur silently.

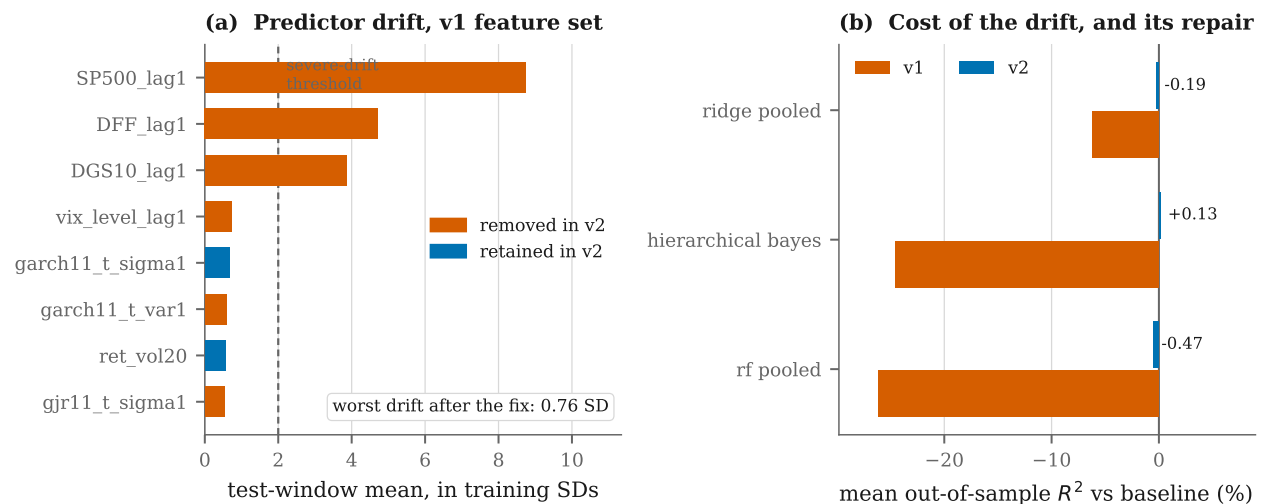


Figure 1: Predictor drift and its repair. Panel (a) shows where the test window sits relative to training moments for the v1 feature set, in training standard deviations, coloured by whether the corrected specification retains the feature. Panel (b) shows the mean out-of-sample  $R^2$  across the eight tickers before and after the correction.

Table 1: Feature dictionary. Every model input is declared here; a column is a feature because it is listed, not because it is numeric. Stationarity is from augmented Dickey–Fuller tests. All retained features are observable at the close of  $t$ , and the target is the return from  $t$  to  $t + 1$ .

| Feature           | Family         | Status   | Note                                                                                                                               |
|-------------------|----------------|----------|------------------------------------------------------------------------------------------------------------------------------------|
| ret               | own return     | Retained | Own log return realised on day $t$                                                                                                 |
| ret_lag1          | own return     | Retained | Own log return on day $t-1$                                                                                                        |
| ret_lag2          | own return     | Retained | Own log return on day $t-2$                                                                                                        |
| ret_lag3          | own return     | Retained | Own log return on day $t-3$                                                                                                        |
| ret_lag5          | own return     | Retained | Own log return on day $t-5$                                                                                                        |
| ret_lag10         | own return     | Retained | Own log return on day $t-10$                                                                                                       |
| ret_vol5          | own volatility | Retained | Trailing 5-day SD of own returns through $t$                                                                                       |
| ret_vol10         | own volatility | Retained | Trailing 10-day SD of own returns through $t$                                                                                      |
| ret_vol20         | own volatility | Retained | Trailing 20-day SD of own returns through $t$                                                                                      |
| mkt_ret_lag1      | market         | Retained | Market proxy (SPY) log return, most recent close                                                                                   |
| vix_log_lag1      | uncertainty    | Retained | Log of the CBOE VIX at the most recent close                                                                                       |
| vix_ret_lag1      | uncertainty    | Retained | Log change in VIX at the most recent close                                                                                         |
| DFF_diff_lag1     | macro          | Retained | Daily change in the effective federal funds rate                                                                                   |
| DGS10_diff_lag1   | macro          | Retained | Daily change in the 10-year Treasury yield                                                                                         |
| garch11_t_sigma1  | garch          | Retained | GARCH(1,1)- $t$ one-step-ahead conditional SD for $t+1$                                                                            |
| SP500_lag1        | macro          | Excluded | Non-stationary level (ADF $p=0.99$ ). Test-window mean sits at $z=+8.73$ under training moments and 100% of test rows...           |
| DFF_lag1          | macro          | Excluded | Non-stationary level (ADF $p=0.83$ ). Test-window mean at $z=+4.72$ , 100% of test rows beyond 3 training SDs. Replaced...         |
| DGS10_lag1        | macro          | Excluded | Non-stationary level (ADF $p=0.80$ ). Test-window mean at $z=+3.86$ , 85% of test rows beyond 3 training SDs. Replaced...          |
| vix_level_lag1    | uncertainty    | Excluded | Stationary, but superseded by its log transform <code>vix_log_lag1</code> .                                                        |
| ret_t             | own return     | Excluded | Bit-identical to ‘ret’ (max absolute difference 0.0). Exact duplicate.                                                             |
| sp500_ret_lag1    | market         | Excluded | Correlates 0.9985 with <code>mkt_ret_lag1</code> ; the two measure the same quantity. Retaining both inflates collinearity and...  |
| garch11_t_var1    | garch          | Excluded | Square of <code>garch11_t_sigma1</code> ; redundant.                                                                               |
| gjr11_t_sigma1    | garch          | Excluded | Held back for the volatility-model robustness check rather than entered alongside <code>garch11_t_sigma1</code> , with which it... |
| gjr11_t_var1      | garch          | Excluded | Square of <code>gjr11_t_sigma1</code> ; redundant.                                                                                 |
| egarch11_t_sigma1 | garch          | Excluded | Training-split fit diverged: values reach $1e+148$ and the training SD is infinite. Excluded pending re-estimation.                |

Table 2: Out-of-sample  $R^2$  (%) against the per-ticker mean baseline, test window 2023–2025,  $n = 751$  per ticker. No entry is significant after multiplicity correction across the 48 model–ticker Diebold–Mariano tests.

| Ticker      | AR(1), per ticker | Zero  | Hierarchical Bayes (t) | Hierarchical Bayes (t) | Random forest, pooled |
|-------------|-------------------|-------|------------------------|------------------------|-----------------------|
| AAPL        | 0.24              | −0.38 | −0.73                  | −0.40                  | −0.64                 |
| AMZN        | −0.00             | −0.45 | −0.10                  | 0.12                   | −0.19                 |
| GOOGL       | 0.01              | −0.47 | −0.23                  | 0.08                   | −1.31                 |
| META        | 0.08              | −0.46 | 0.07                   | 0.40                   | −0.73                 |
| MSFT        | −1.56             | −0.42 | −0.27                  | −0.07                  | −0.55                 |
| NVDA        | 0.37              | −1.00 | −0.25                  | 0.72                   | −0.09                 |
| ORCL        | 0.01              | −0.12 | 0.05                   | 0.03                   | −0.02                 |
| TSLA        | 0.02              | −0.15 | −0.03                  | 0.21                   | −0.23                 |
| <b>Mean</b> | −0.10             | −0.43 | −0.19                  | 0.13                   | −0.47                 |

Table 3: Volatility forecast comparison on an identical sample of 6008 test observations, scored by QLIKE against the squared return.  $p$ -values are Diebold–Mariano against the rolling-variance benchmark with a Newey–West HAC variance. \*\*\*, \*\*, \* denote 1%, 5% and 10%.

| Specification          | QLIKE   | MSE vs proxy          | DM $p$    |
|------------------------|---------|-----------------------|-----------|
| GARCH(1,1)- $t$        | −6.5361 | $5.56 \times 10^{-6}$ | 0.0000*** |
| EGARCH(1,1)- $t$       | −6.5100 | $5.38 \times 10^{-6}$ | 0.0009*** |
| GJR-GARCH(1,1)- $t$    | −6.5035 | $5.44 \times 10^{-6}$ | 0.0005*** |
| Rolling variance (20d) | −6.4000 | $5.66 \times 10^{-6}$ | —         |

Table 4: Probabilistic accuracy and interval calibration, 6008 test observations. Log score and CRPS are means; coverage is empirical against the nominal level.  $p$ -values are Diebold–Mariano on the log-score differential against the hierarchical model.

| Model                                  | Log score | CRPS     | 50%   | 80%   | 90%   | 95%   | DM $p$    |
|----------------------------------------|-----------|----------|-------|-------|-------|-------|-----------|
| Hierarchical Bayes (t)                 | 2.4777    | 0.012 14 | 0.536 | 0.825 | 0.918 | 0.958 | —         |
| Hierarchical Bayes (t), SD calibration | 2.4757    | 0.012 15 | 0.545 | 0.833 | 0.923 | 0.960 | 0.0000*** |
| Hierarchical Bayes (t), uncalibrated   | 2.4728    | 0.012 13 | 0.462 | 0.757 | 0.873 | 0.932 | 0.0606*   |
| Ridge + GARCH scale                    | 2.3516    | 0.012 20 | 0.572 | 0.834 | 0.909 | 0.943 | 0.0000*** |
| Zero mean + GARCH scale                | 2.3496    | 0.012 22 | 0.571 | 0.833 | 0.908 | 0.944 | 0.0000*** |
| Zero mean + constant variance          | 2.1968    | 0.012 46 | 0.548 | 0.798 | 0.869 | 0.910 | 0.0000*** |
| <i>Nominal</i>                         |           |          | 0.500 | 0.800 | 0.900 | 0.950 |           |

Table 5: Cross-asset transfer, averaged over the eight held-out tickers. At  $k = 0$  the held-out ticker’s intercept is integrated over its population distribution; at  $k > 0$  its last  $k$  training days are revealed and the model refitted.

| $k$ days | RMSE     | Log score | CRPS     | Coverage 90% |
|----------|----------|-----------|----------|--------------|
| 0        | 0.023 62 | 2.472 67  | 0.012 14 | 0.871 84     |
| 20       | 0.023 62 | 2.472 40  | 0.012 14 | 0.871 84     |
| 60       | 0.023 62 | 2.472 35  | 0.012 14 | 0.871 50     |
| 120      | 0.023 62 | 2.472 20  | 0.012 14 | 0.872 00     |
| 250      | 0.023 62 | 2.472 21  | 0.012 14 | 0.871 50     |

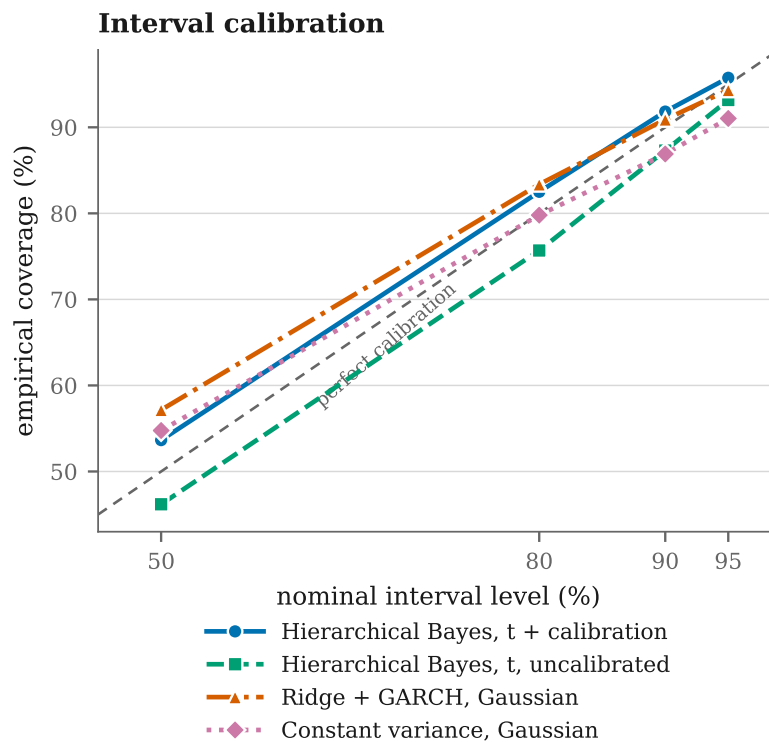


Figure 2: Interval calibration: empirical coverage against nominal level for four predictive specifications, test window,  $n = 6008$ .



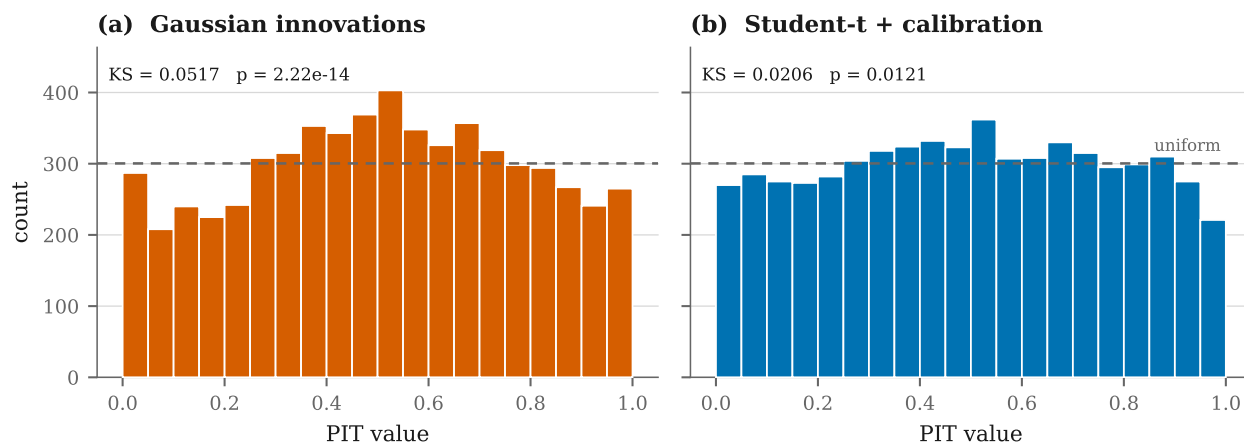


Figure 3: Probability integral transform histograms. Under a correctly specified predictive distribution these are uniform; the dashed line marks the uniform density and each panel reports a Kolmogorov–Smirnov test against it.

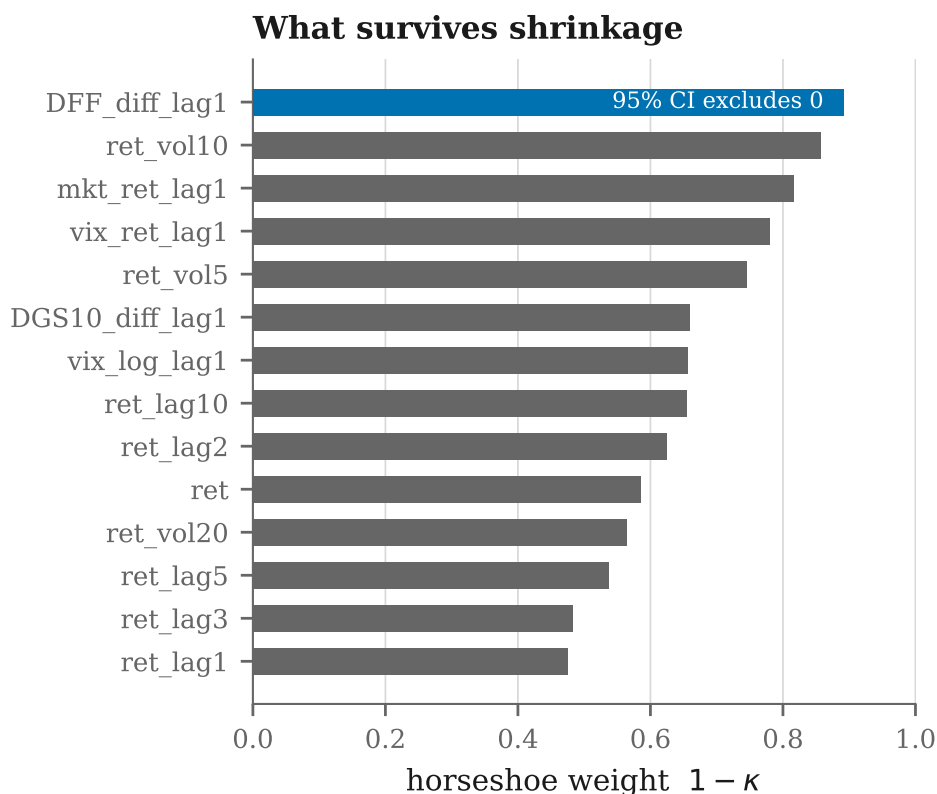


Figure 4: Horseshoe shrinkage weight  $1 - \kappa$  by predictor. Values near one indicate that the data overrode the prior. The single highlighted predictor has a 95% credible interval excluding zero.

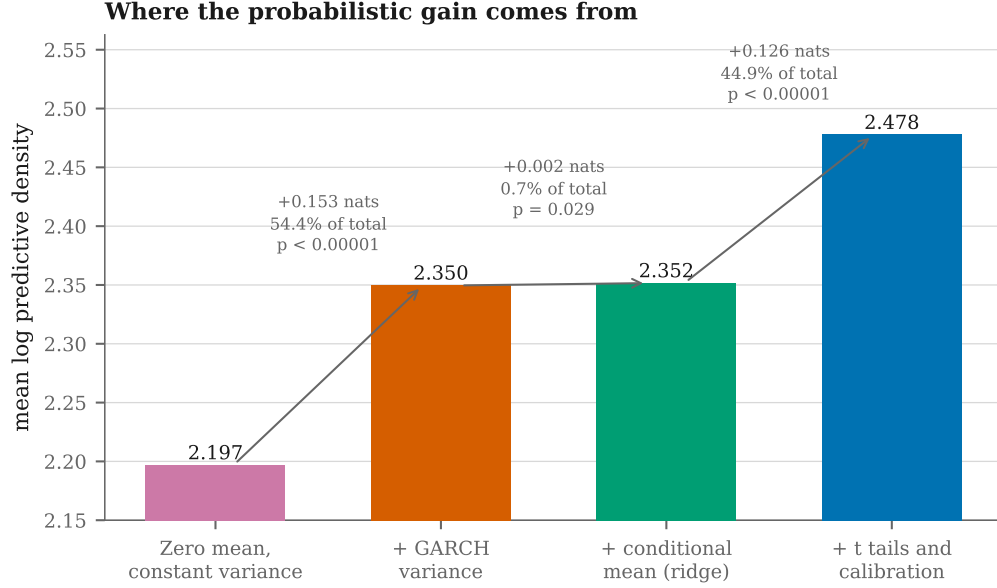


Figure 5: Decomposition of the gain in mean log predictive density over a null forecast. Each increment is labelled with its size in nats, its share of the total gain, and the Diebold–Mariano  $p$ -value against the preceding rung.

## Data Availability Statement

Price data are from Yahoo Finance and macro-financial series from the Federal Reserve Economic Data service, both publicly available. All code, processed data, audit scripts and pinned dependencies required to reproduce every number, table and figure in this paper are available at [https://github.com/RichelCode/Final\\_Stock\\_Analysis](https://github.com/RichelCode/Final_Stock_Analysis).

## Conflict of Interest

The authors declare no conflict of interest.

## Author Contributions

**Author 1:** Conceptualization, Data curation, Methodology, Formal analysis, Writing – original draft. **Author 2:** Data curation, Methodology, Investigation, Writing – original draft. **Author 3:** Supervision, Project administration, Validation, Writing – review and editing.

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